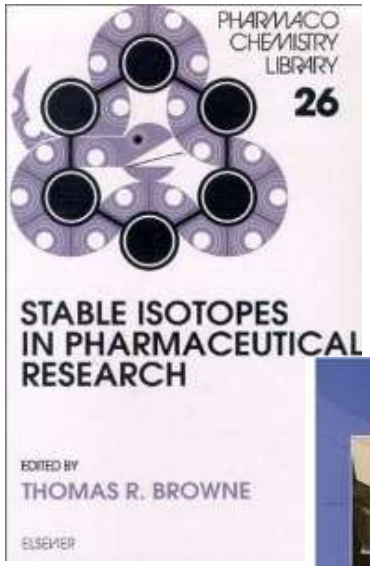
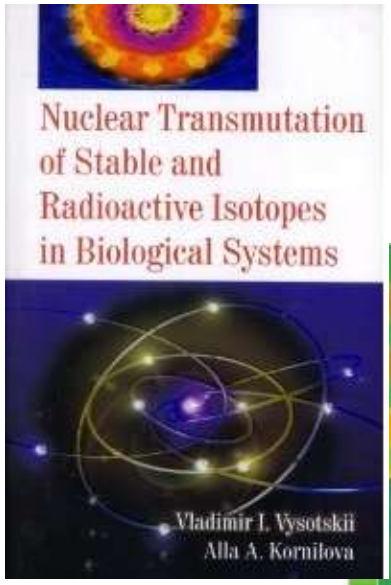
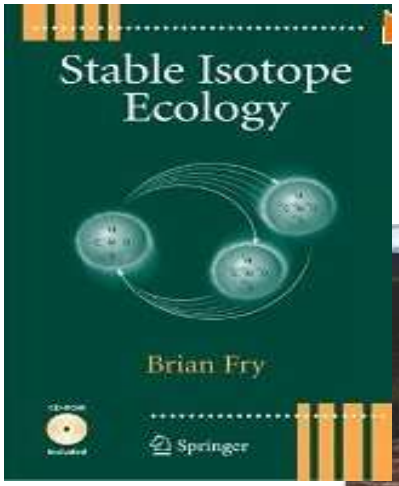
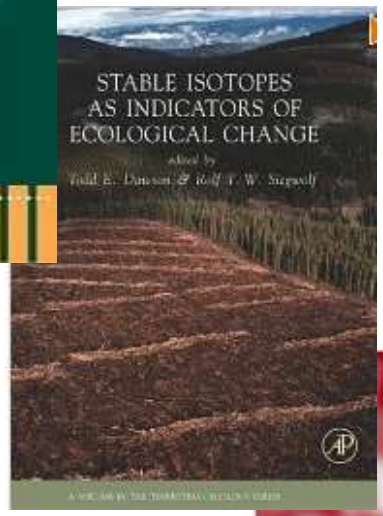
Stable Isotopes

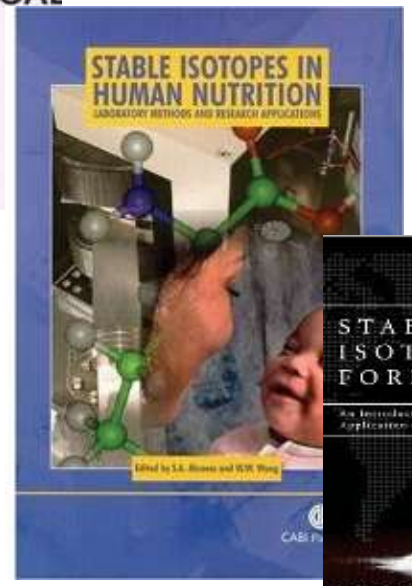


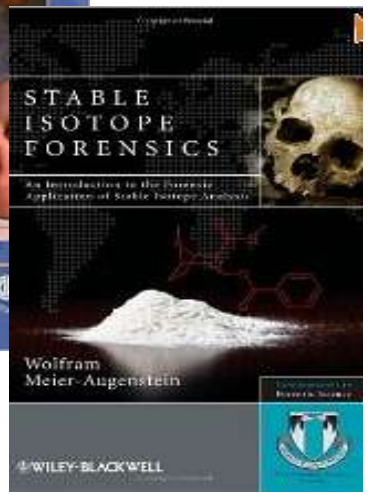


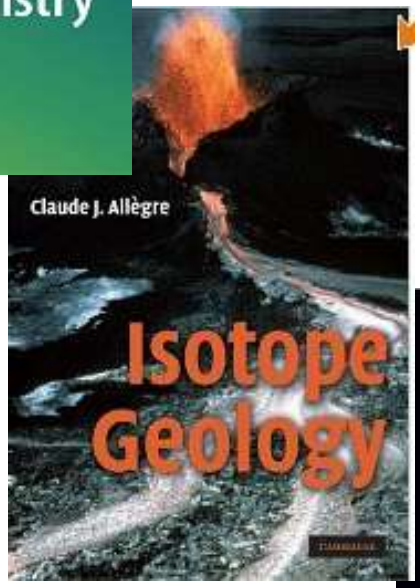


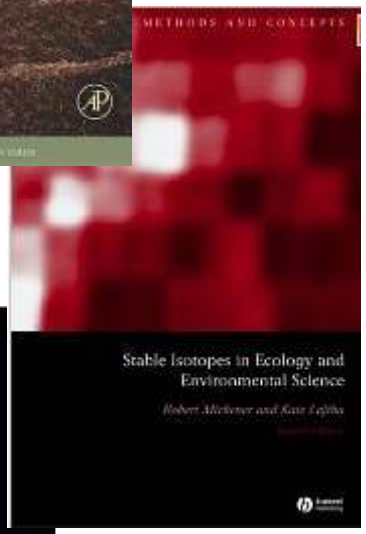


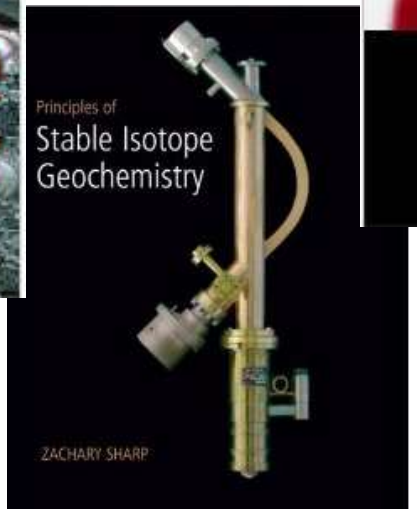












Applications in Earth and Environmental Sciences

Thermometry

- *formation T of rocks, minerals and gases*
- *based upon T dependence of isotope fractionation*

Palaeoclimatology

- *Isotope ratios of minerals, fluids and gases preserve information about environmental conditions (most notably temperature) of the past.*

Tracers

- *large terrestrial reservoirs (ocean, mantle, meteoric waters, organic matter) have characteristic isotope signatures*
- *trace the origin of rocks, fluids, plants, contaminants etc.*

Processes

- *diffusion, crystallisation, open and closed systems, bacterial processes*

Also biological, archaeological, medical and chemical sciences

Isotope Fractionation

- *Fractionation* refers to the partial separation of two isotopes of the same element, producing reservoirs with different ratios of the isotopes. These differences are called isotopic *anomalies*.
- Two classes of basic mechanisms and one strange one exist for fractionating isotopes:
  - The two basic mechanisms are:
    - *equilibrium isotope fractionation,* which is due to differences in bond *energies* of isotopes in compounds
    - *kinetic isotope fractionation*, which is due to differences in average *velocity* or *reaction rates* of different isotopes
    - Both depend only on the *mass* of the isotope and are called *mass dependent fractionation*; both will fractionate, say $^{18}O/^{16}O$ about twice as much as $^{17}O/^{16}O$
  - The strange mechanism is *mass-independent fractionation and depends on molecule symmetry*

Notation

*We can define a **fractionation factor** (α):*

$$\alpha_{A-B} = \frac{R_A}{R_B}$$

Where R_A, R_B are the isotope ratios in two phases (ex. carbonate and water, or water vapor and water, etc)

<u>NOTE</u>: α is close to 1 because ratios differ by parts per thousand

We report isotopic fractionations as δ ("delta") values in "per mil" units:

$$\delta^{18}\mathrm{O} = \left(\frac{\mathrm{R}_{\mathrm{sample}}}{\mathrm{R}_{\mathrm{s\,tan\,dard}}} - 1 \right) \times 1000$$

Note that 'deltas' are named after the heavy isotope

$$\delta_a - \delta_b \sim 1000 \ln\alpha$$

*So each isotopic measurement is reported relative to a **standard***

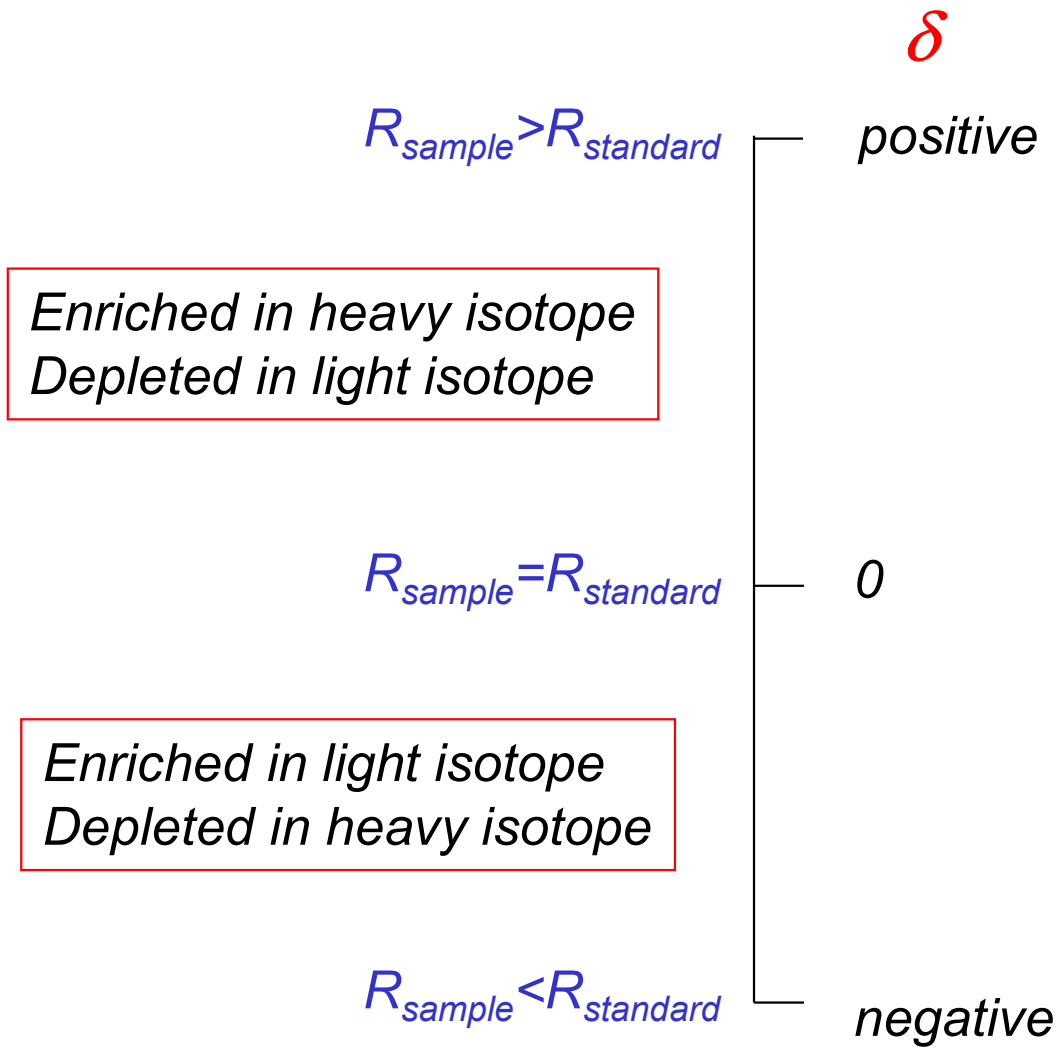
Table 9.1. Isotope Ratios of Stable Isotopes

Element	Notation	Ratio	Standard	Absolute Ratio
Hydrogen	δD	D/H ($^2H/^1H$)	SMOW	1.557×10^{-4}
Lithium	δ^6Li	$^6li/^7Li$	NBS L-SVEC	0.08306
Boron	$\delta^{11}B$	$^{11}B/^{10}B$	NBS 951	4.044
Carbon	$\delta^{13}C$	$^{13}C/^{12}C$	PDB	1.122×10^{-2}
Nitrogen	$\delta^{15}N$	$^{15}N/^{14}N$	atmosphere	3.613×10^{-3}
Oxygen	$\delta^{18}O$	$^{18}O/^{16}O$	SMOW, PDB	2.0052×10^{-3}
	$\delta^{17}O$	$^{17}O/^{16}O$	SMOW	3.76×10^{-4}
Sulfur	$\delta^{34}S$	$^{34}S/^{32}S$	CDT	4.43×10^{-2}

Delta scale

1. *Isotope ratios "calibrated" using a standard which has an isotopic ratio that is internationally agreed.*

2. *The "delta" is in units of per mil because whole numbers are easier for us to understand and remember.*



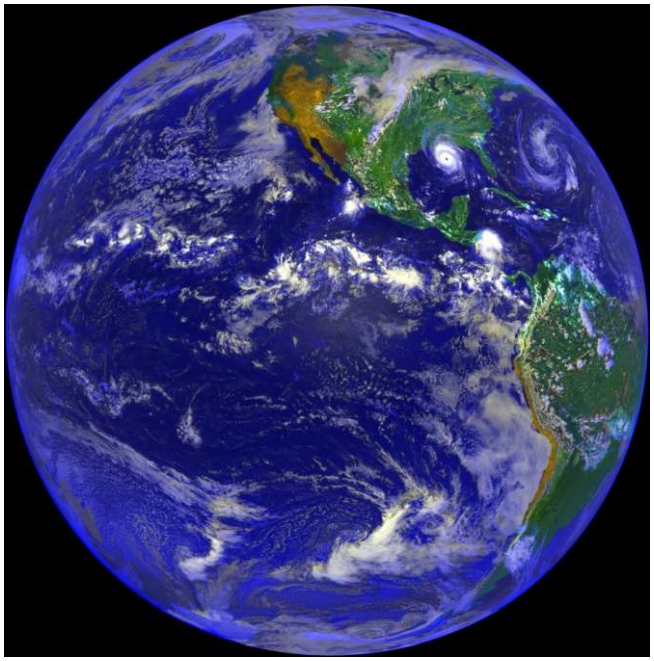
Fractionation of O and H between water and vapour

Standard mean ocean water (SMOW) $\delta D = 0‰$ and $\delta^{18}O = 0‰$

$H_2{}^{16}O$ has higher vapour pressure than $H_2{}^{18}O$ or D-bearing water ("light isotopes move faster")

During evaporation or condensation the vapour is enriched in ^{16}O and H and the liquid is enriched in ^{18}O and D

Much more fractionation between hydrogen and deuterium than oxygen isotopes due to relative mass differences difference
Oxygen: (18-16)/16 = 1/8.
Hydrogen: (2-1)/1=1



Fractionation factors at 25°C

$$\alpha_{L-V} = \frac{\left(^{18}O/^{16}O\right)_L}{\left(^{18}O/^{16}O\right)_V} = 1.0092$$

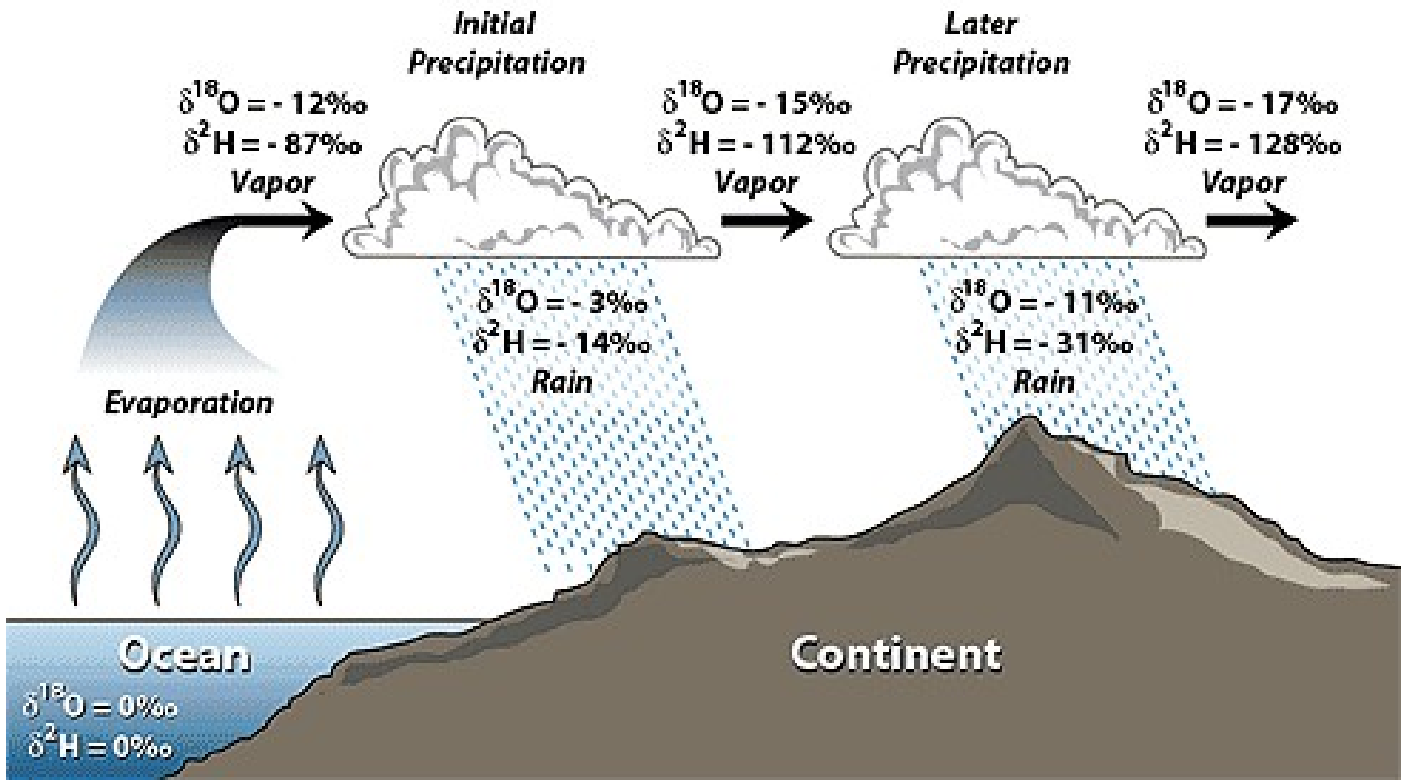
$$\alpha_{L-V} = \frac{(D/H)_L}{(D/H)_V} = 1.074$$

Isotopic changes associated with precipitation

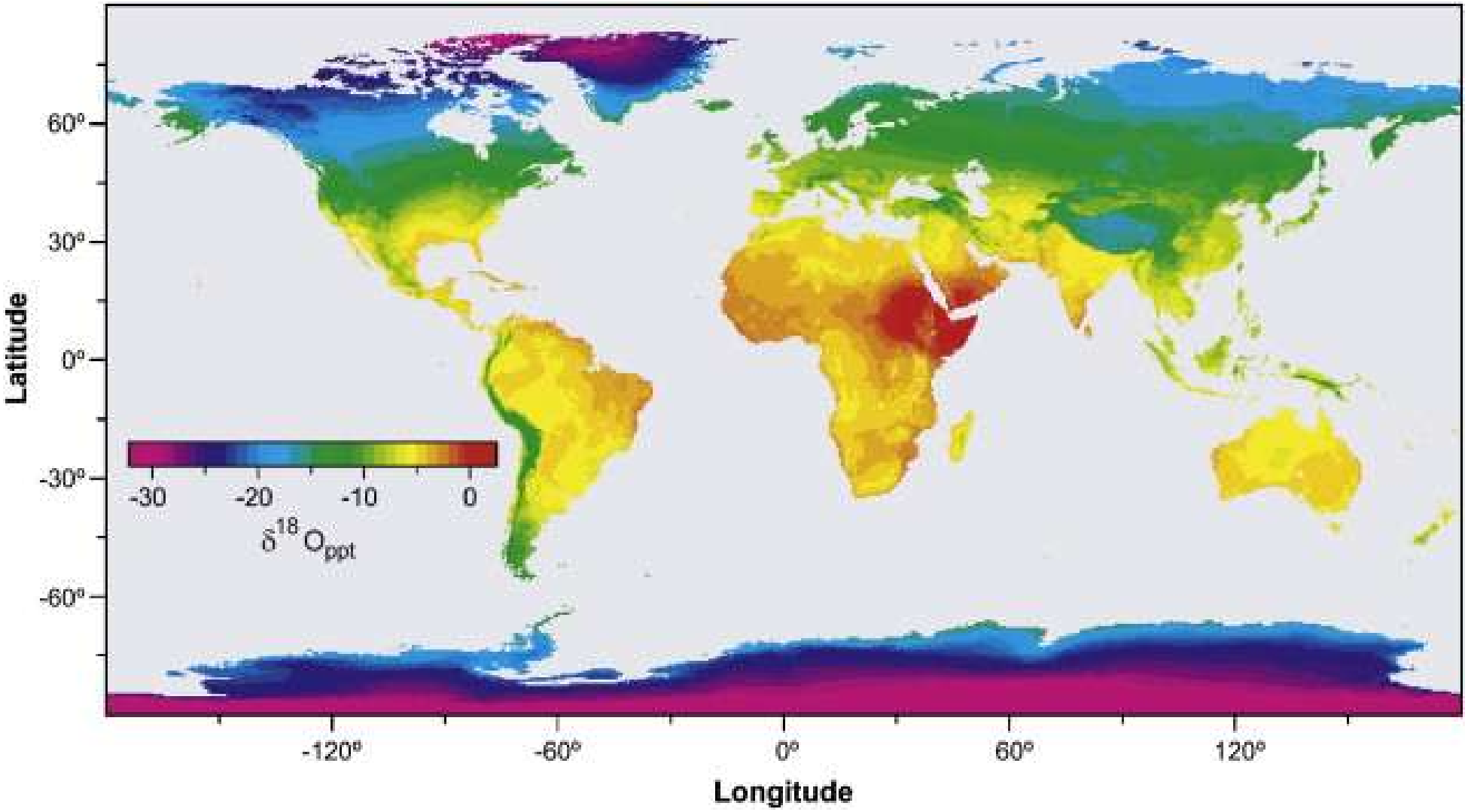
Rayleigh fractionation:

$$\frac{R_A}{R_0} = f^{(\alpha-1)}$$

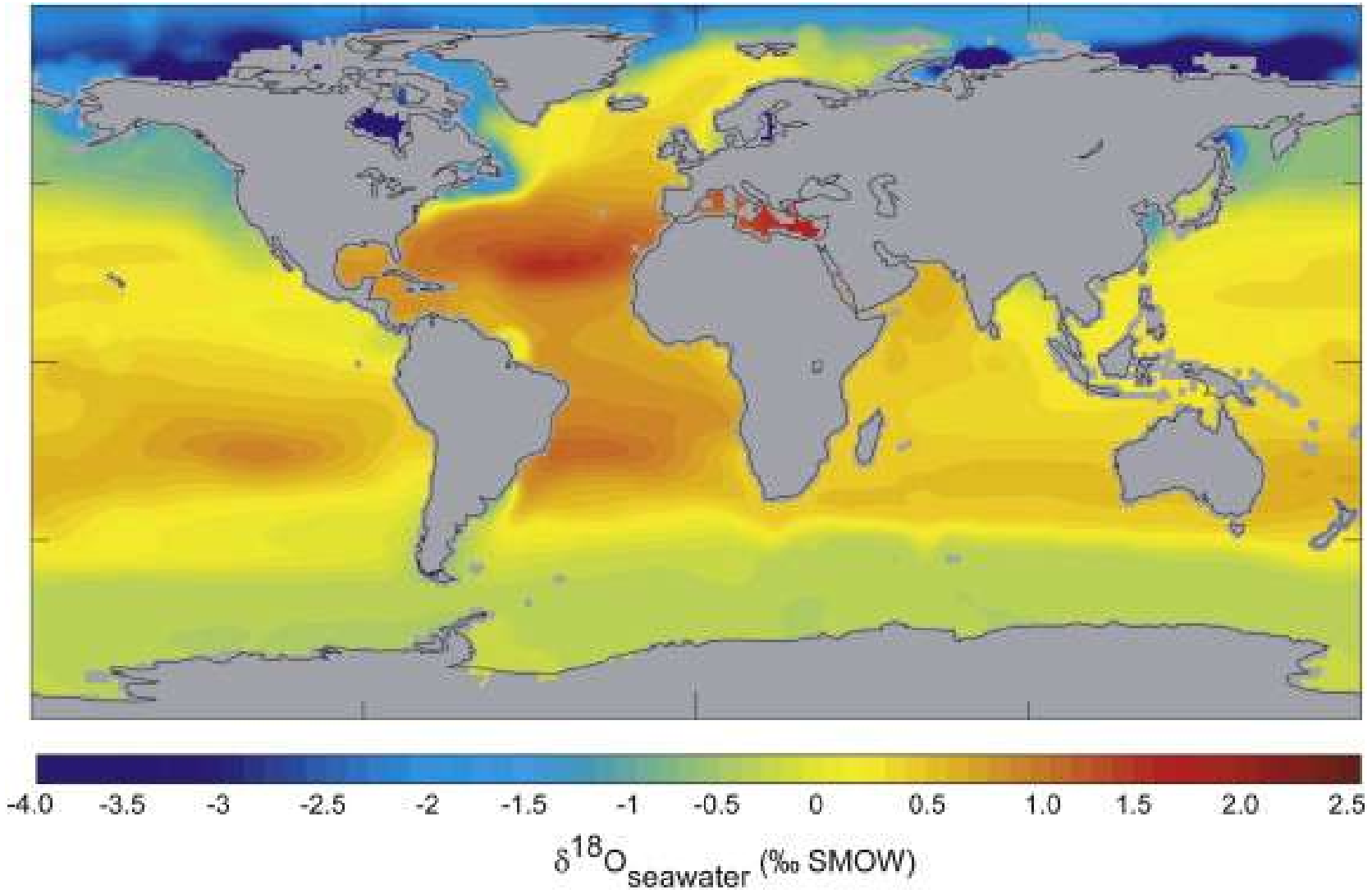
If the tropics are the source of all cloud moisture, then the $\delta^{18}O$ of rainfall increases from equator to pole.



Oxygen isotopic composition of rainfall

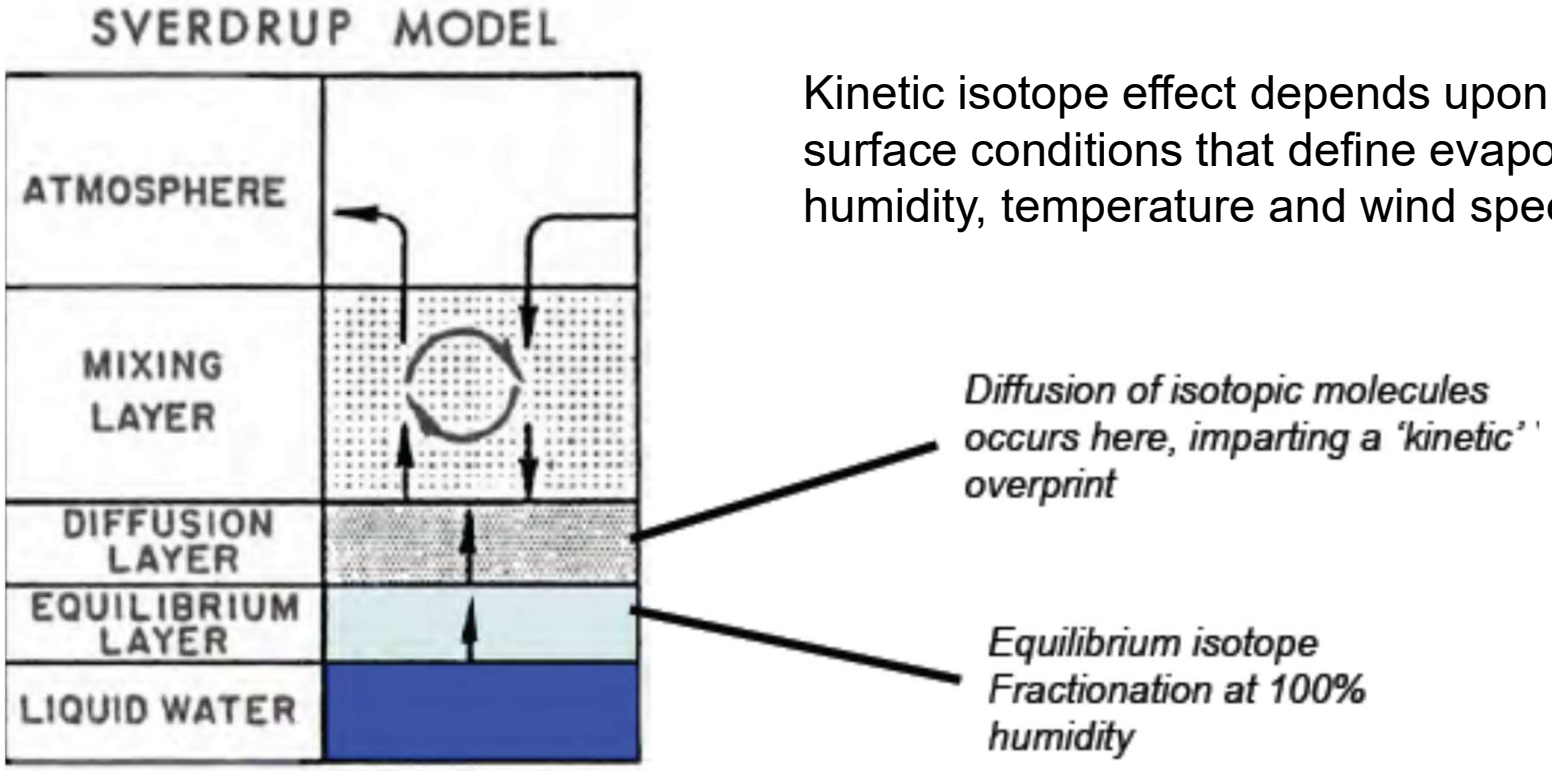


Oxygen isotopic composition of seawater



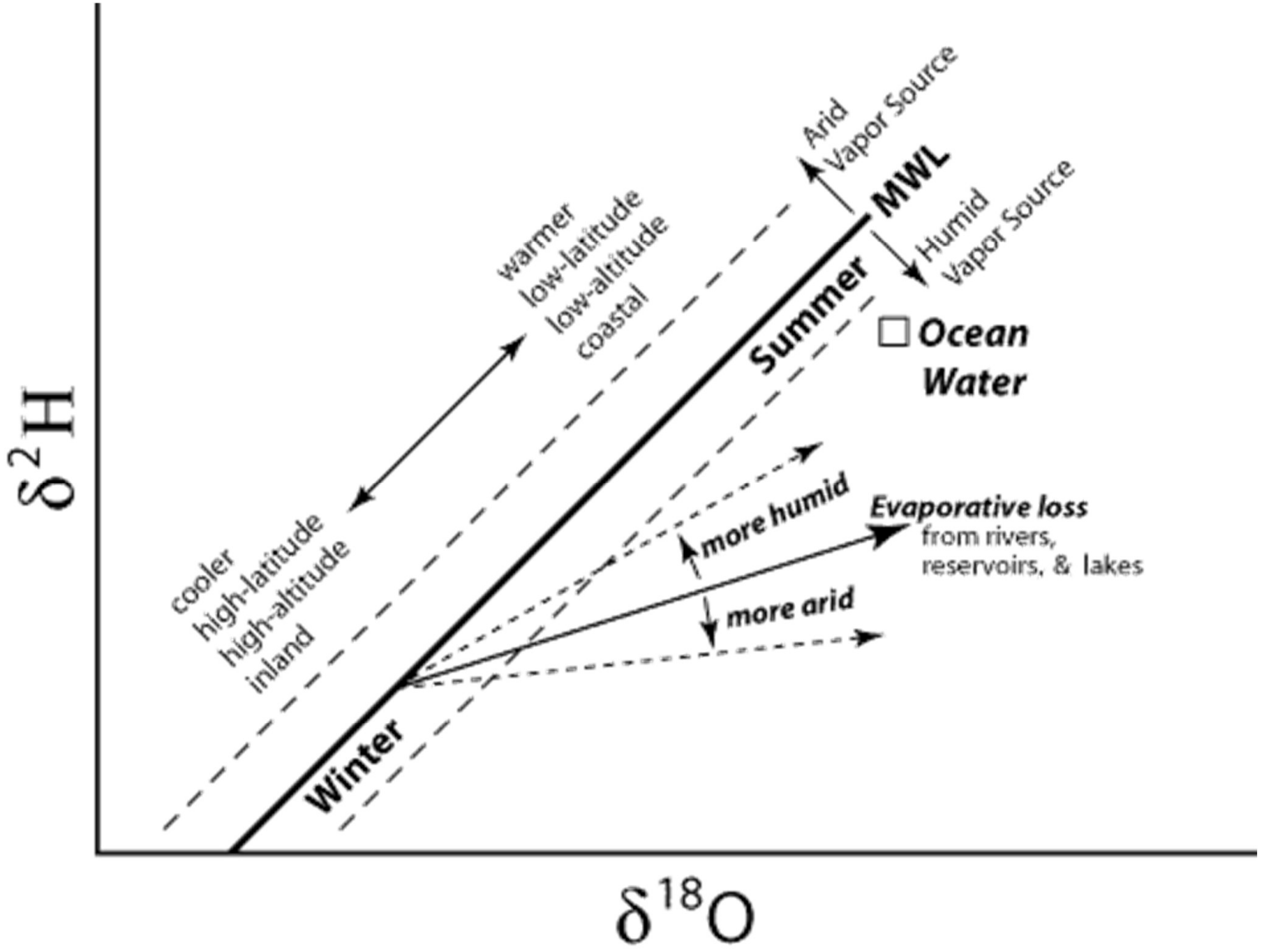
Kinetic isotope fractionation effects

At low humidity the maximum kinetic isotope fractionation



Kinetic isotope effect depends upon the sea surface conditions that define evaporation i.e. humidity, temperature and wind speed

Summary of fractionation processes

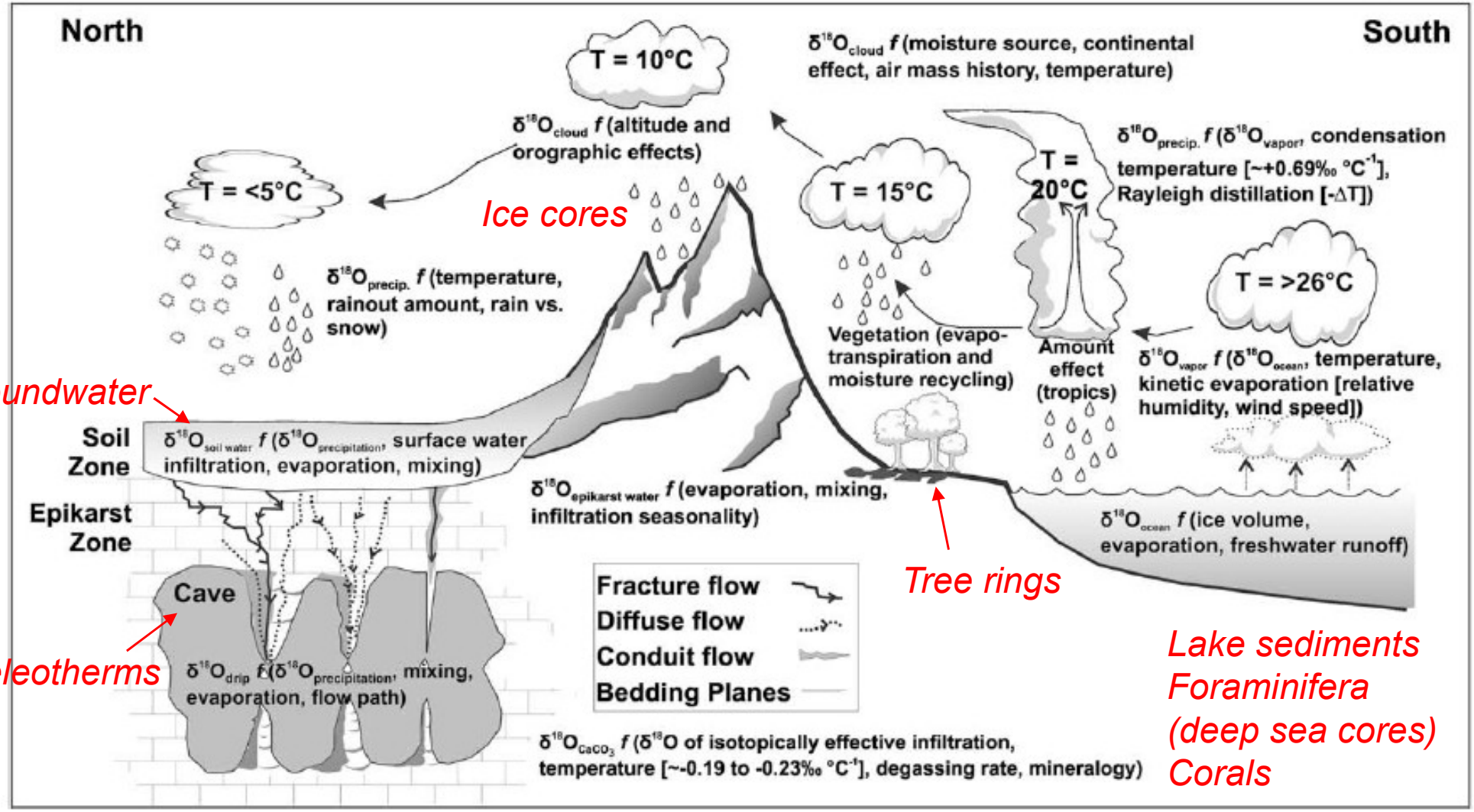


Palaeoclimatology and ocean water isotope evolution



The Glomar Challenger built in the 1960's was the first ship specifically designed for the purpose of drilling into and taking core samples from the deep ocean floor

Processes fractionating isotopes in the hydrosphere



Lachniet, 2009

Carbonate $\delta^{18}O$ – temperature relationships

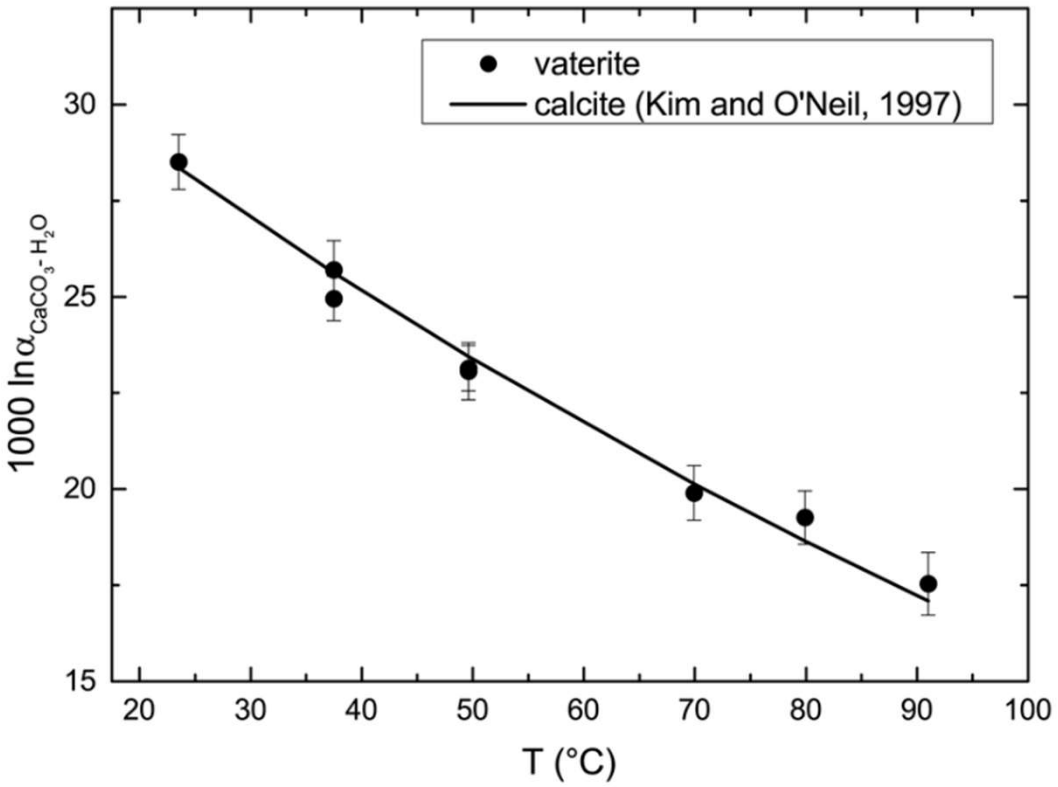
The relationship between water-$\delta^{18}O$, temperature, and the equilibrium $\delta^{18}O$ of calcite was determined **empirically** by Epstein et al., (1953) and later modified by Craig (1965) – ^{18}O enriched in carbonate.

Relationship described by:

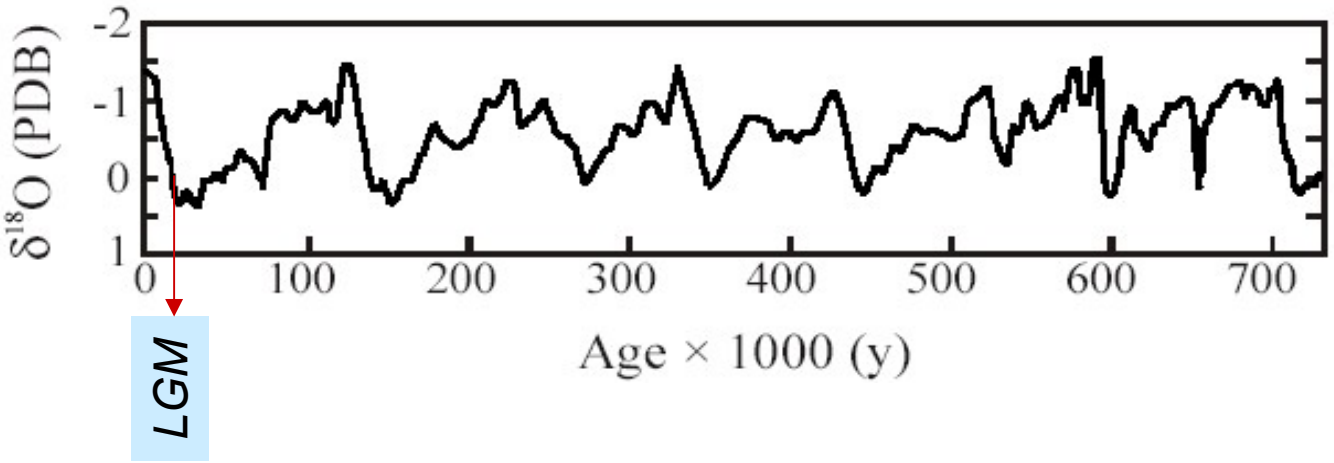
$1000 \cdot \ln\alpha = 17599/T - 29.64$

Remember $\delta_a - \delta_b \sim 1000 \ln\alpha$ therefore:

1000 ln (α) vs T can be read as the difference between the δ values as a function of T

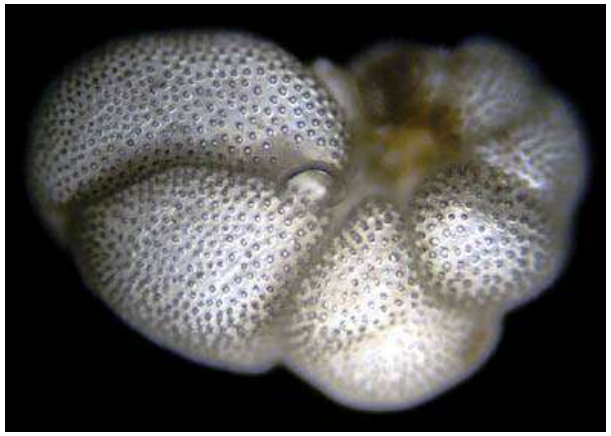


Glacial-Interglacial foraminifera $\delta^{18}O$



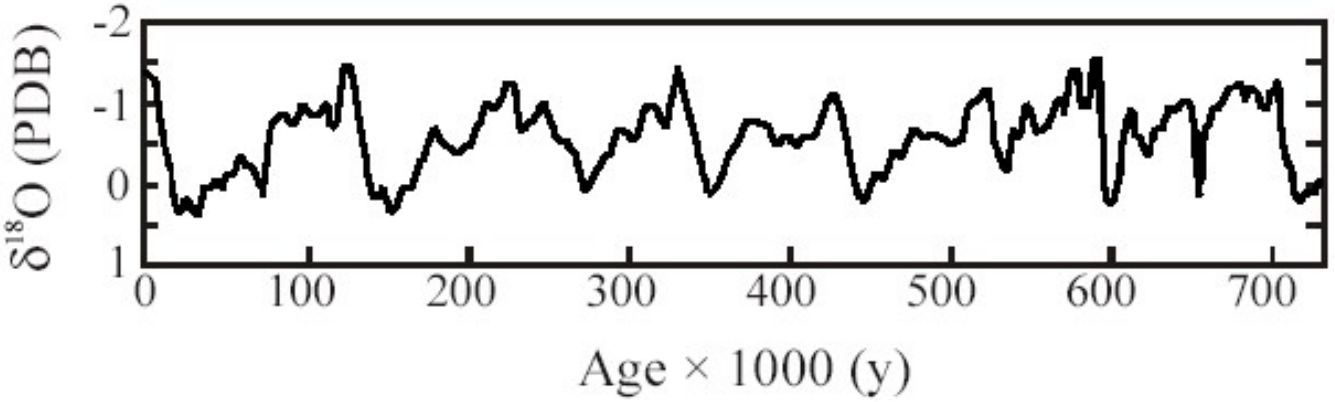
1. The patterns are saw-toothed, implying that glaciation is a slow process and that deglaciation occurs much more rapidly.
2. A periodicity of ~100,000 years in the patterns can be reasonably linked to one of the Milankovich periodicities in orbital forcing.
3. There are many more glacial/interglacial stages in the Pleistocene than thought previously.

deep-sea foraminifera



SPECMAP – Spectral Mapping Project - benthic $\delta^{18}O$ record, used to date marine sediments of unknown age – global stratigraphic correlation tool

Foraminifera $\delta^{18}O$ records



Data from deep-sea (benthic) foraminifera show +1.5‰ $\delta^{18}O$ shift during Last Glacial Maximum

Foraminifera will record different temperatures depending upon the depth in the ocean where they live – they can be planktonic (marine floaters) or benthic (sea floor dwellers) they will record temperatures that are a few deg-C different.
Planktonic and benthic forams can easily be distinguished, because specific forams only live in special conditions and environments.

Foraminiferal calcite can recrystallise in sediments, changing the original $\delta^{18}O$ value.
We can overcome this problem by carefully selecting shells that do not show signs of alteration.

Additional problems might arise from mixing of shells of different ages during bioturbation, which is especially problematic in areas with low sedimentation rates.
We could overcome this by choosing a site with a high sedimentation rate.

Water isotopes and glacial cycles

The "Ice Volume" effect
Light isotope removed from ocean, locked into large ice sheets.

Ocean $\delta^{18}O$ shift recorded in marine carbonates that grew during glacial period.

Key question:
How much of the shift is due to light isotopes locked up in ice sheets and how much is temperature dependent fractionation?

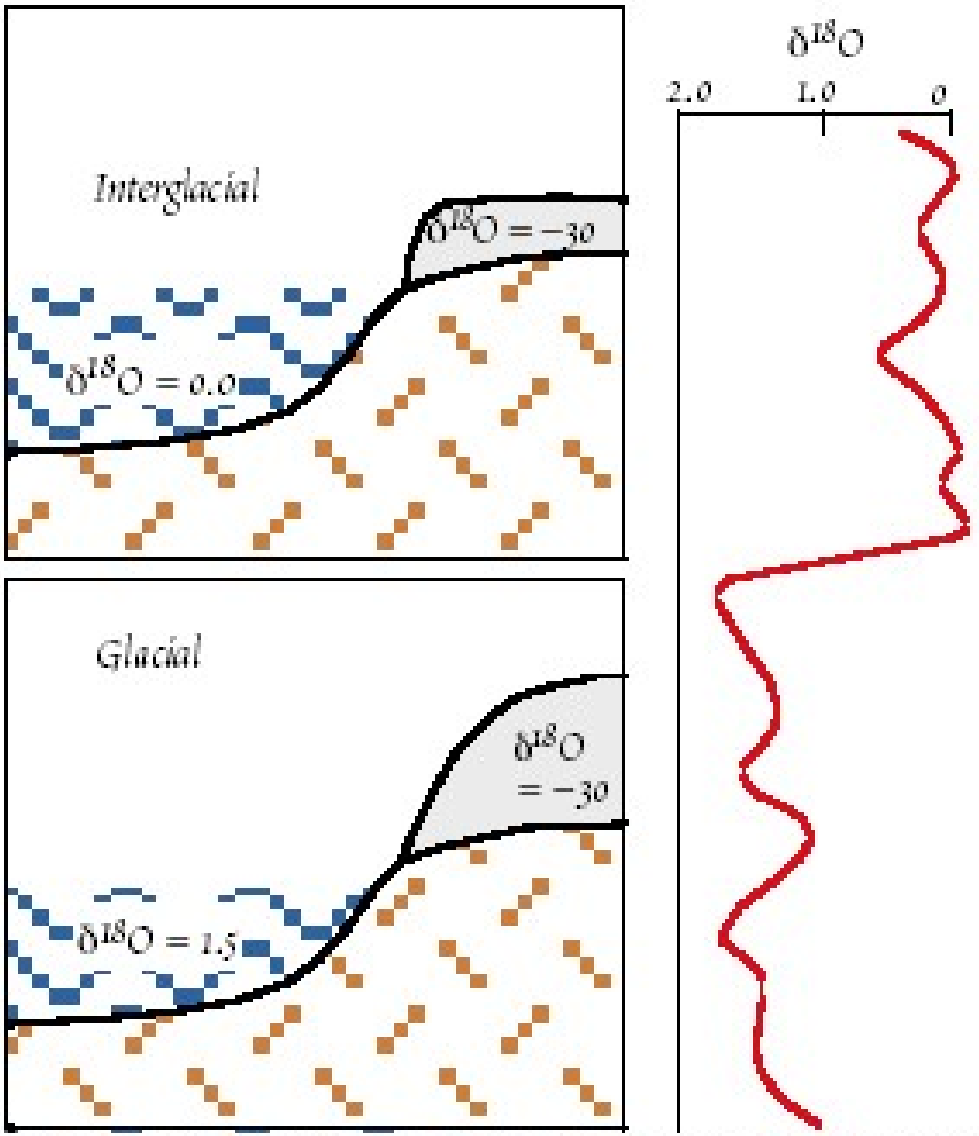


Figure 35.2. Cartoon illustrating how $\delta^{18}O$ of the ocean changes between glacial and interglacial periods.

Lachniet, 2009

Pore fluids – glacial ocean water $\delta^{18}O$

Attempts to quantify the glacial-interglacial change in seawater $\delta^{18}O$; Result = +0.7-0.8‰

Need to model pore fluid profiles because:

advection drives water out of compacting sediments

&

diffusion works to erase pulse Signal

only works for last glacial maximum

Atlantic ocean

glacial enrichment

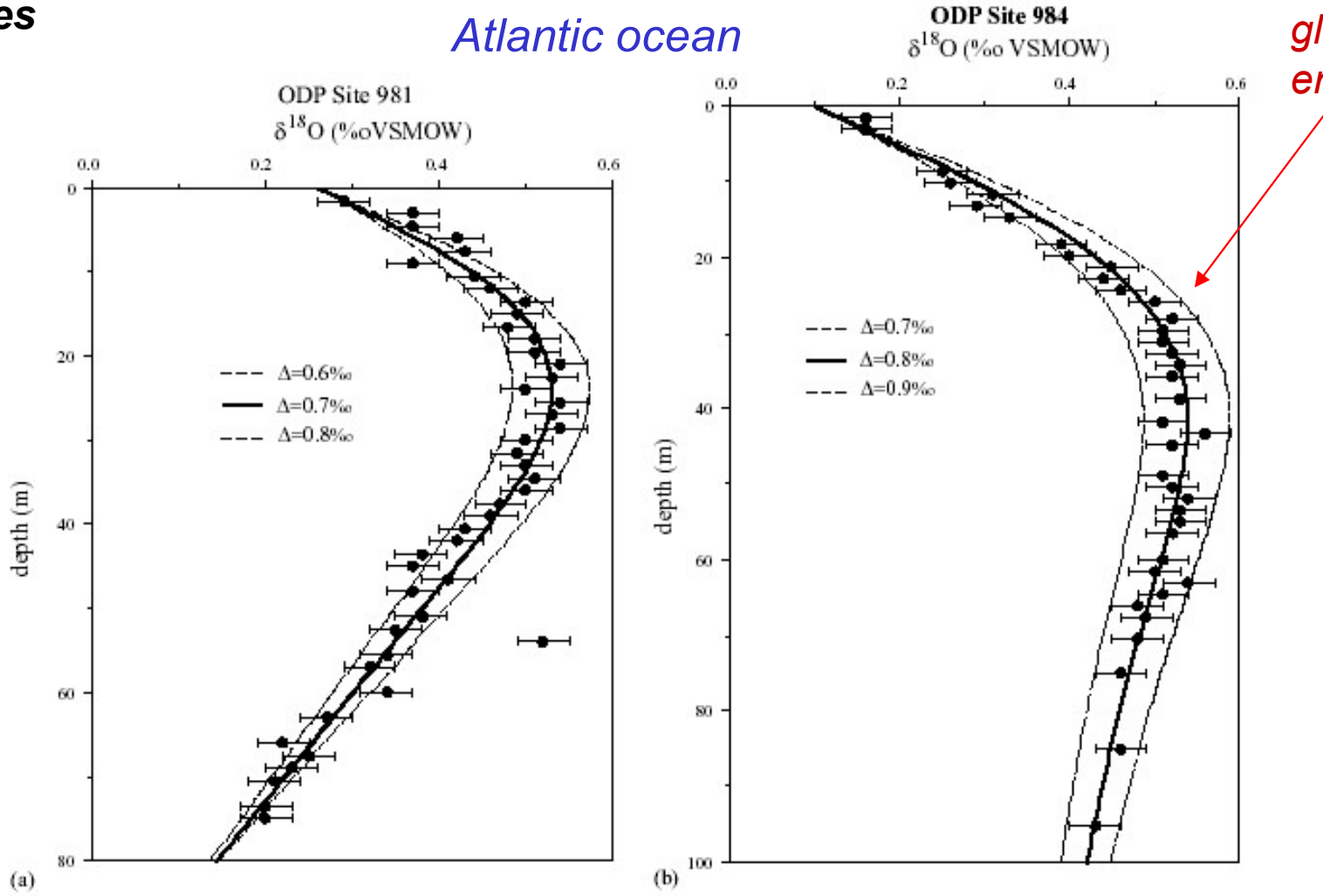
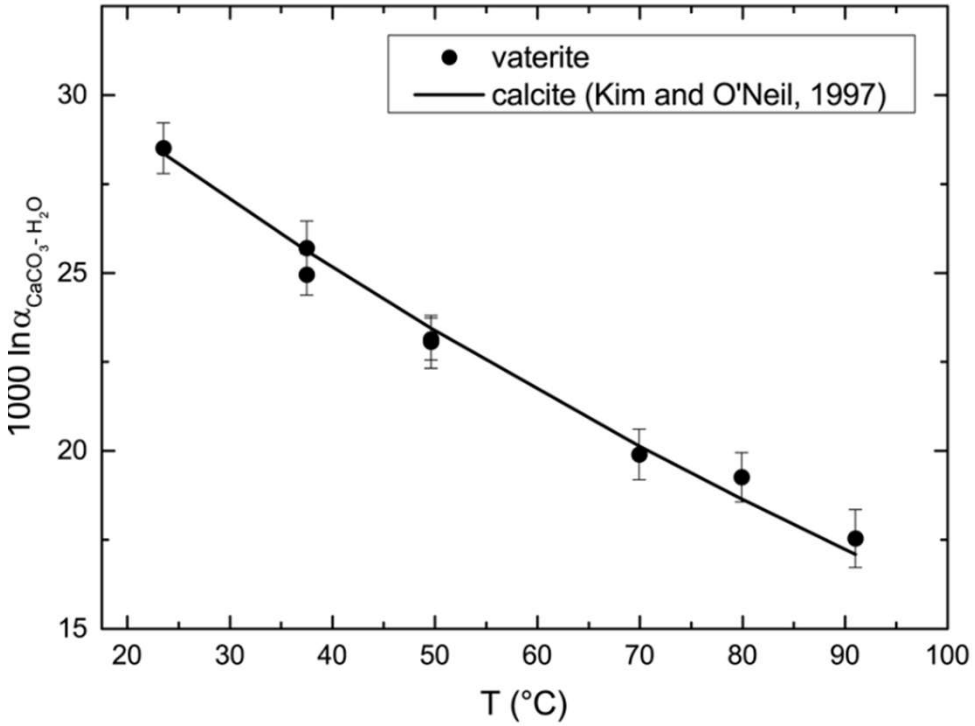


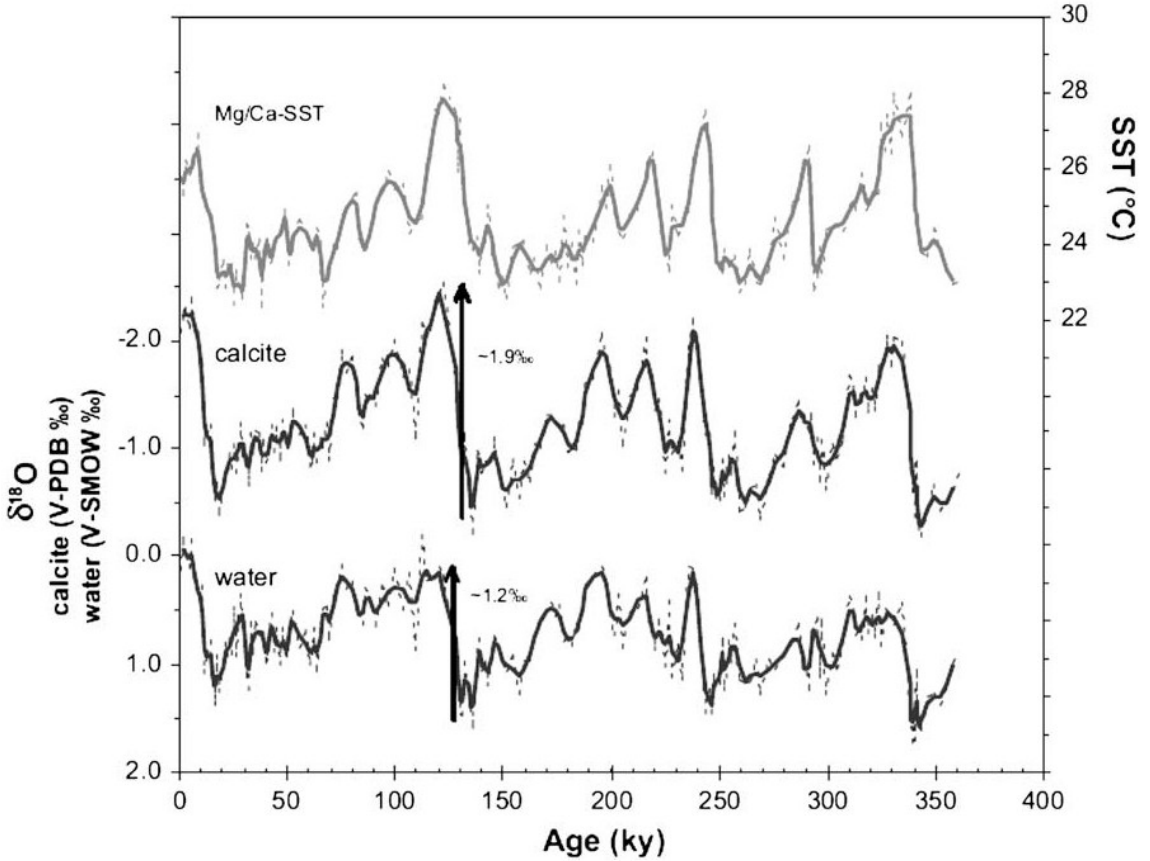
Fig. 2. Oxygen isotope data on pore fluids from ODP Leg 162, Sites 981 (A) and 984 (B). The lines indicate the model results for different changes in the $\delta^{18}O$ of seawater.

Summary of the really big question in paleoceanography:
How much of this shift was due to ice volume (sea level change) and how much was due to temperature change?

Schrag modelled the glacial-interglacial shift in porewater $\delta^{18}O$ (~0.7-0.8‰), so we have 0.7-0.8‰ left over for temperature change if 1.5‰ in forams is correct.

0.75‰ equates to 4°C: (0.75/(15*80) from figure below = 4)





Clumped isotopes

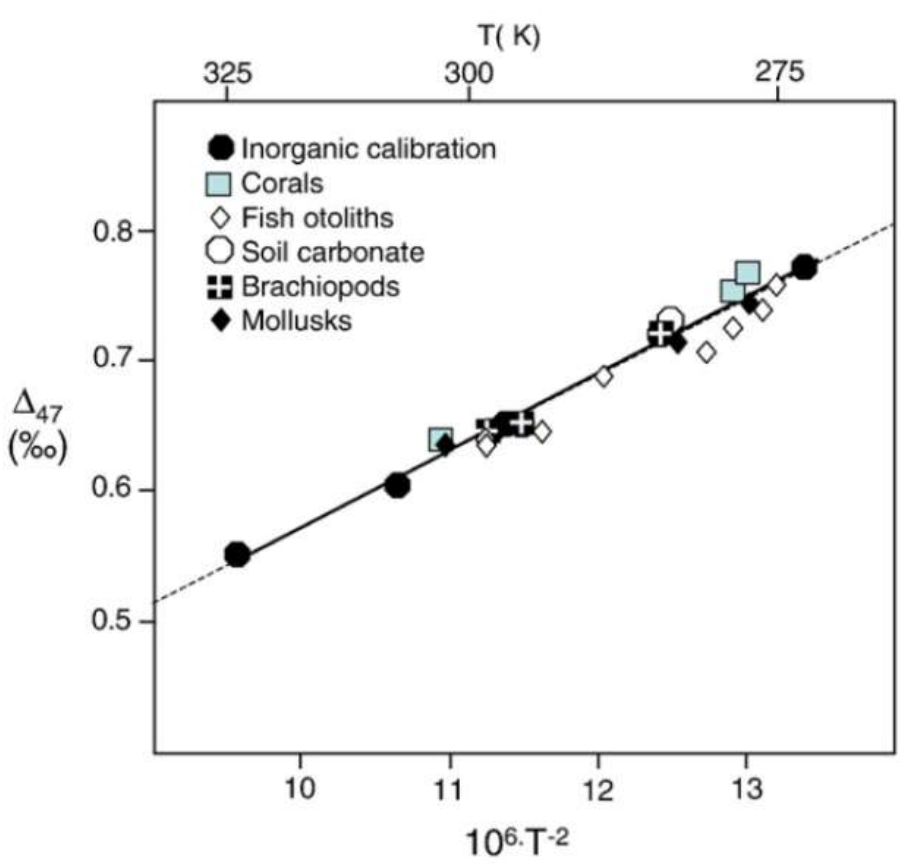
- *What are rare isotopologues or clumped isotopes?*
  - *molecules composed of more than one rare isotope*

$$M^{12}C^{18}O^{16}O_2 + M^{13}C^{16}O_3 = M^{13}C^{18}O^{16}O_2 + M^{12}C^{16}O_3$$

- *Why are they useful?*
  - *^{13}C-^{18}O bond is more abundant than it should be. This is T dependent hence a thermometer independent of source composition.*
  - *Intra-phase or intra-molecular isotope signal that can be read without reference to any other species*

Carbonate clumped-isotope thermometry

$$Ca^{13}C^{16}O^{16}O^{16}O + Ca^{12}C^{18}O^{16}O^{16}O \rightleftharpoons Ca^{13}C^{18}O^{16}O^{16}O + Ca^{12}C^{16}O^{16}O^{16}O$$



H_3PO_4

$^{13}C^{18}O^{16}O$ (mass 47)

$$\Delta_{47} = \left(\frac{(47/44)_{actual}}{(47/44)_{random}} - 1 \right) \times 1000$$

Mass 47--- $^{13}C^{18}O^{16}O$ (mostly)
Mass 44--- $^{12}C^{16}O^{16}O$

Slide from W. Guo

Some specific science applications

Intra-phase or intra-molecular isotope signal that can be read without reference to any other species (i.e. source fluid).

— *Palaeoclimate – forams, coccoliths, concretions, speleothems, dinosaurs, paleosols*

— *Dinosaur blood temp 36-38°C*

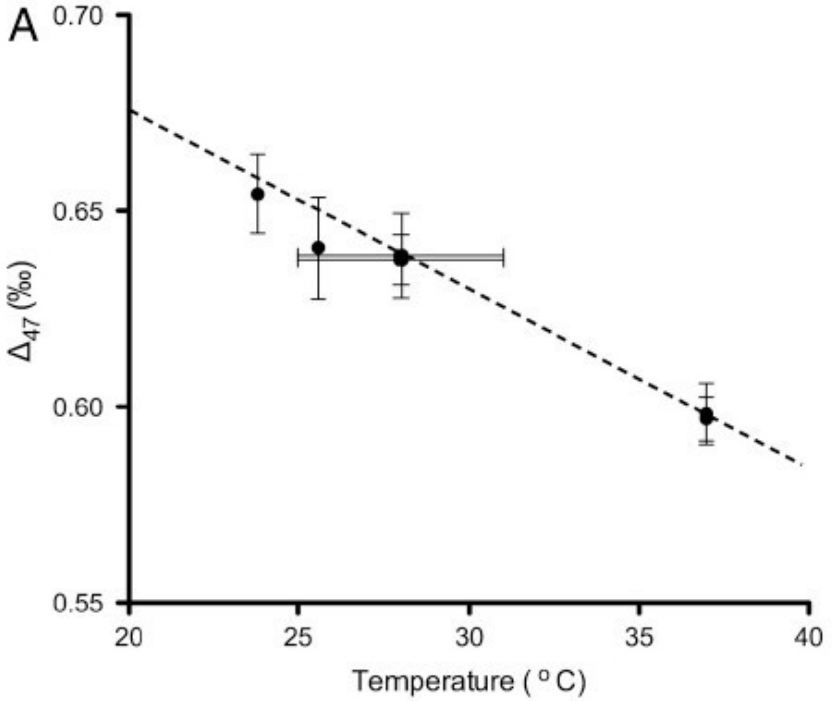


Table 1. Δ_{47} measurements from tooth bioapatite of modern species

Species	n *	Δ_{47}	Estimated Body Temperature (°C) †	Δ_{47} Temperature (°C) ‡
White Rhinoceros (*Ceratotherium simum*)	6	0.597 ± 0.006	37	36.6 ± 1.4
Indian Elephant (*Elephas maximus indicus*)	4	0.596 ± 0.008	37	36.9 ± 2.0
Nile Crocodile (*Crocodylus niloticus*)	7	0.638 ± 0.006	28 ± 3	26.9 ± 1.4
American Alligator (*Alligator mississippiensis*)	3	0.639 ± 0.011	28 ± 3	26.7 ± 2.5
Sand Tiger Shark (*Carcharias taurus*)	5	0.641 ± 0.013	25.3	26.2 ± 3.0
Sand Tiger Shark (*Carcharias taurus*)	3	0.654 ± 0.010	23.6	23.1 ± 2.3

Summary

Learning outcomes for today:

- Understand isotope fractionation and temperature dependence
- Understand chemical and physical fractionation
- Fractionation and the global hydrologic cycle – the meteoric water line
- Controls on the hydrologic cycle
- Methods of palaeoclimate reconstruction
  - how we use shells + complications
  - How we use pore fluids + complications
- How we can use clumped isotopes for palaeoclimate

Background reading

Bill White's on-line isotope geochemistry textbook:

Part 4: Stable Isotope Geochemistry

Chapters 26 and 27

http://www.geo.cornell.edu/geology/classes/Geo656/656notes03.html

Lachniet M.S. (2009) Climatic and environmental controls on speleothem oxygen-isotope values. Quaternary Science Reviews, 28,412–432

International Atomic Energy Authority: Environmental Isotopes in the Hydrological Cycle – Volume II

- downloaded from http://www-naweb.iaea.org/napc/ih/IHS_resources_publication_hydroCycle_en.html

Gat J.R. (1996) Oxygen and hydrogen isotopes in the hydrological cycle. Annual Reviews of Earth and Planetary Science 24, 225-262.